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Course:- ITA0502
code:- Computer vision
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CO2-AT3 Test

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1A) * Image enhancement in the spatial domain means Improving the quality of an Image by directly the Pixel values

Theechniques:-

a) Gray-level :- Transformations

* Changes the brightness and Contrast of an Image.
* Operates directly on Pixel Intensity.

Ex:-

* Simple and fast.
* Improves brightness,

Applications:-

* Medical Images
* Satellite Images.
* Dark Photographs

b) Histogram Processing.

* A Histogram shows how Pixel intensities are distributed.

Techniques:-

* Histogram equalization.
* Histogram Matching.

→ Applications:-

* x-ray Images
* low-contrast Photographs

c) Spatial filtering:-

* uses a small mask (kernel) to Process Pixels.

Types:-

smoothing filters:-
* mean filter
* Gaussian filter
* median filter.

[sh]arpening filters:-
* laplacian filter
* High-Boost filter.

[I]mage smoothing vs Image smoothing:-

Image smoothing:-
[re]moves noise
[-] low pass filter
[b]lurs image.

[I]mage Sh[arpening]

Applications:-

* Smoothing: noise removal.
* SharPening: Edge detection and object recognition.

3) Image Enhancement in frequency Domain.

A) * Enhances images After converting them into the frequency domain using fourier transform.

→ low-Pass filter (LPF).

* Removes noise.
* Smooths the Image.
* Used in Medical Imaging.
* used in LPF- smoothing.

→ High-Pass filter (HPF)

* SharPness edges.
* Enhances details.
* Used in Satellite and industrial Images,

* Difference:-

* LPF → smoothing
* HPF → smoothing.

4) Image segmentation:-

* It divides an image into meaningful regions

Techniques:-

→ Point Detection:-

* Detects isolated pixels

→ Line Detection:-

* Detects straight lines

→ Edge Detection:-

* Detects object boundaries

→ Thresholding:-

* Seperates objects by intensity

→ Regions Based Segmentation

→ Applications:-

* Medical Imaging
* face
* Object.

Q5) Edge Detection and Edge Linking and Boundary Detection:-

1) → Edge Detection:-

* finds Object edges using Sobel, Prewitt or Canny Operators

→ Edge Linking:-

* Connects broken edges to form Continuous boundaries

Image

↓

Edge Detection

↓

Edge Linking

↓

Boundary Detection

↓

Object Recognition